

Information Technology University, Lahore

Applied Machine Learning BSCE- 23 Spring-2026 Assignment # 01

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BSCE23042

Question # 01: Linear Regression with Custom Dataset

A custom dataset named Marketing_Sales was generated with 500 records.

- **Independent Variables (X):** TV_Budget, Social_Media_Budget, and Newspaper_Budget.
- **Dependent Variable (Y):** Sales.

--- First 5 Rows of Marketing_Sales (Analyst: SF) ---

	Campaign_ID	TV_Budget	Social_Media_Budget	Newspaper_Budget	Sales
0	SF_001	218.543053	142.650726	22.587628	46.831348
1	SF_002	477.821438	111.858310	56.480590	64.172259
2	SF_003	379.397274	68.810247	87.929854	44.455862
3	SF_004	319.396318	164.621054	74.561364	58.056422
4	SF_005	120.208388	140.098923	81.623309	49.089599

X_train shape: (350, 3)

X_test shape: (150, 3)

--- Model Results for Roll No: 42 ---

Intercept: 14.2744

Coefficients (TV, Social, News): [0.051223 0.18190599 0.01332557]

R2 Score: 0.9668

MAE: 1.7269

MSE: 4.8812

--- Actual vs Predicted Sales (First 10 for SF) ---

	Actual	Predicted
361	50.690788	49.482593
73	64.149646	59.041205
374	21.122754	21.552568
155	27.250898	29.663053
104	45.179877	45.069901
394	49.733787	50.093974
377	41.832492	42.683229
124	44.673355	46.172942
68	54.214266	52.138263

450 57.262176 58.019303

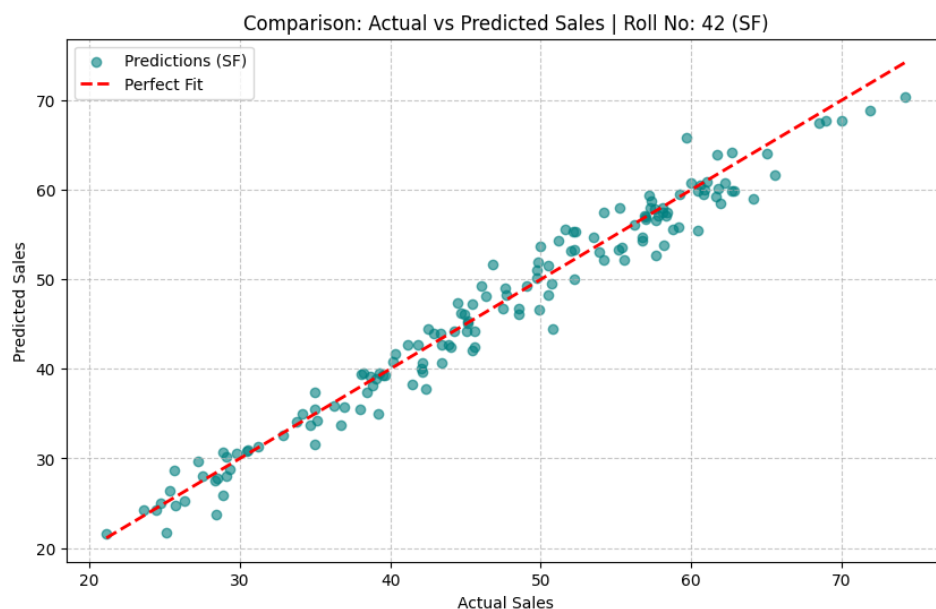
2. Model Results & Evaluation

The model was built using the Normal Equation for Multiple Linear Regression.

- **Intercept:** 14.2744
- **Coefficients:** TV: 0.0512
 - **Social Media:** 0.1819
 - **Newspaper:** 0.0133
- **Performance Metrics:**
 - **R² Score:** 0.9668 (indicates the model explains 96.68% of sales variance).
 - **MAE:** 1.7269
 - **MSE:** 4.8812

3. Interpretation

- **Best Channel:** **Social Media** contributes most to sales with the highest coefficient (0.1819).
- **Practical Meaning:** For every \$1 unit increase in Social Media budget, sales are expected to increase by approximately 0.18 units, assuming other budgets remain constant.



Question # 02:

1. Data Preparation & Exploration

The Iris dataset consists of 150 samples. To ensure a robust evaluation, the data was shuffled and split into:

- Training Set: 70% (used to teach the models).
- Testing Set: 30% (used to evaluate performance on unseen data).
- Features: Sepal Length, Sepal Width, Petal Length, Petal Width.

2. Model Performance Summary

The table below compares the three algorithms based on their accuracy on the test dataset.

Algorithm	Hyperparameters	Accuracy Score
K-Nearest Neighbors (KNN)	k=3,5,7	100.00%
Logistic Regression	Max Iterations: 1000	93.33%
Decision Tree	Max Depth: 3	95.56%
Export to Sheets		

3. Evaluation Metrics & Visuals

K-Nearest Neighbors (KNN)

- Best Performance: Achieved perfect classification (1.00 Precision/Recall).
- Logic: It classifies a flower based on the "majority vote" of its nearest neighbors in the feature space.
- Confusion Matrix: A clean diagonal with zero off-diagonal values, meaning no misclassifications.

Logistic Regression

- Accuracy: 93.33%.
- Observation: This model struggled slightly with the overlap between *Versicolor* and *Virginica*, which are more similar to each other than to *Setosa*.
- Logic: It uses a logistic function to model a binary (or multinomial) outcome.

Decision Tree

- Accuracy: 95.56%.
- Logic: It breaks down the data into a series of "If-Then" rules (e.g., *if Petal Width < 0.8, then Setosa*).
- Observation: While highly interpretable, it was slightly less accurate than KNN in this specific test.

4. Comparative Analysis

1. Which algorithm performed best?

KNN was the clear winner for this dataset, achieving 100% accuracy. The physical dimensions of the Iris species are distinct enough that "clustering" by proximity is the most effective way to separate them.

2. Strengths and Weaknesses

- KNN:
 - * *Strength*: Extremely accurate for small, well-defined datasets.
 - *Weakness*: Becomes very slow if the dataset grows to millions of records.
- Logistic Regression:
 - *Strength*: Fast to train and provides probability scores.
 - *Weakness*: Assumes linear boundaries; struggles if the data "wraps" around itself.
- Decision Tree:
 - *Strength*: Very easy for humans to understand and explain.
 - *Weakness*: High risk of "overfitting" (memorizing the training data rather than learning patterns).
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3. Recommendation for Production

I recommend KNN (with k=5). For the Research Institute's current needs, the dataset is small and the priority is maximum accuracy. KNN provides the most reliable classification for these physical measurements. If the institute plans to collect millions of samples in the future, we should then transition to a Decision Tree for better computational efficiency.

Confusion Matrices for All Models - Analyst: SF

